

BESCOM SMART METER INTELLIGENCE & LOSS DETECTION

Project Abstract: Our system is an intelligent non-invasive overlay designed to fortify BESCOM's smart grid. By harmonizing localized demand forecasting with explainable anomaly detection it preempts grid volatility and identifies revenue leakage and transforming raw high-frequency data into auditable, actionable intelligence for a resilient energy future.

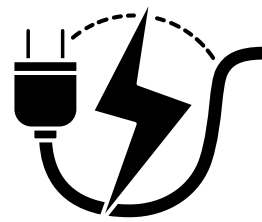
Team Details

Team Name: Team 007

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University: VIT Bhopal

CURRENT OPERATIONAL CHALLENGES:



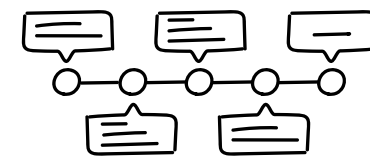
Localized Demand Spikes

Rapid, unpredictable surges in electricity usage at the household or feeder level. Without hourly forecasting, these spikes cause sudden transformer stress, leading to localized blackouts and equipment damage.



Revenue Leakage (NTL)

Non-Technical Losses (NTL) caused by energy theft, physical meter tampering, or bypassing. These irregularities are often indistinguishable from high usage without advanced pattern recognition.



Operational Invisibility

A lack of real-time "eyes" on the distribution network. BESCOM cannot currently distinguish between legitimate consumption variability and suspicious behavior, leading to inefficient manual inspections..



Data Underutilization

High-frequency (15-min) data is being collected but remains "dark." The system lacks a predictive overlay to transform raw consumption numbers into actionable grid-balancing intelligence.

Smart Meter Analytics & Grid Resilience

Our mission is to deploy an intelligent, non-invasive AI layer that synchronizes BESCOM's high-frequency data with real-time operational needs. By converting millions of raw consumption pulses into predictive insights, we bridge the gap between static billing and dynamic grid management—securing revenue and ensuring stability for Bangalore's power infrastructure.

- **Actionable Intelligence:** Transforming 15-minute raw data into high-confidence inspector cues.
- **Revenue Protection:** Systematically identifying bypass and tampering to eliminate non-technical losses.
- **Proactive Grid Stability:** Forecasting localized demand surges before they trigger feeder-level stress.
- **Trust by Design:** Delivering fully auditable, explainable outputs through a local-inference architecture.

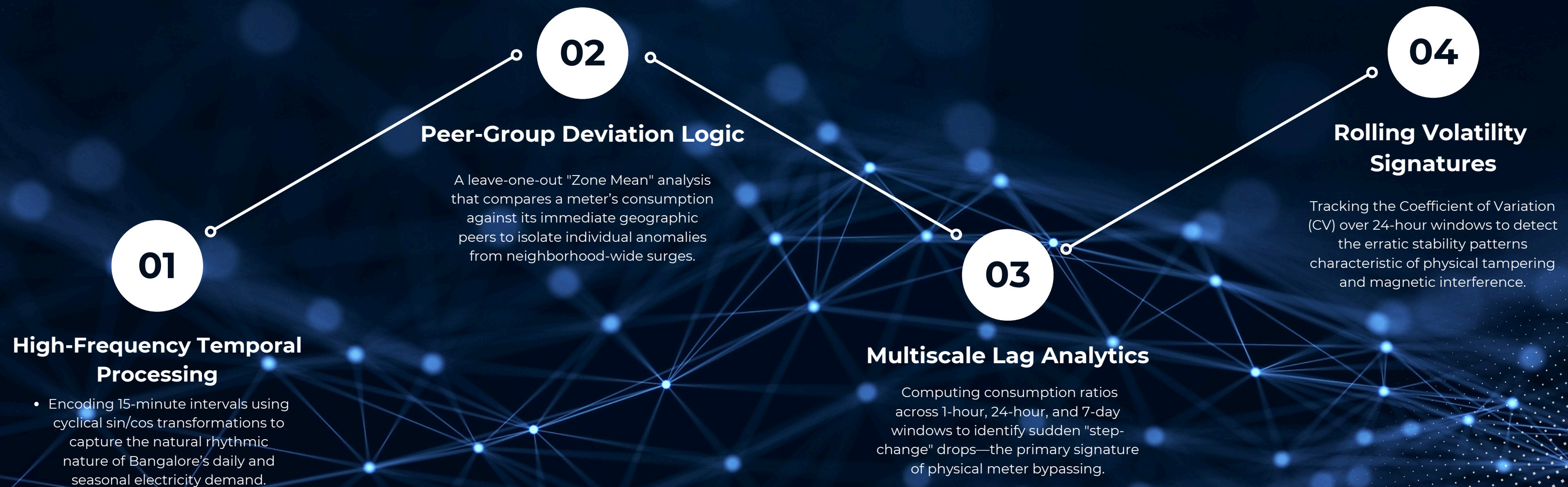
Strategic Solution Architecture

The architecture functions as a passive intelligence layer that extracts value without requiring modifications to existing hardware or core systems. By operating as a "Read-Only" overlay, the solution maintains the integrity of the current Meter Data Management System (MDMS) while providing a sophisticated analytical filter to protect both grid stability and revenue.

- **Zero-Friction Integration:** Consumes standard data exports (CSV/API) from existing infrastructure, requiring no hardware changes or operational downtime.
- **Data Privacy & Sovereignty:** Built on a "Local Inference" model where all AI processing occurs within the secure utility perimeter, ensuring sensitive consumption data never leaves the premises.
- **Decoupled Analytic Engine:** The machine learning pipeline operates independently of the billing system, eliminating the risk of accidental interference with core financial operations.
- **Modular Decision Support:** Acts as a pure decision-support layer, delivering prioritized alerts and evidentiary cues to field teams rather than making autonomous, irreversible changes.



DATA ORCHESTRATION & FEATURE ENGINEERING



Predictive Localized Demand Forecasting

Demand Intelligence (Part A)

Precision in load management begins with granular forecasting. By moving beyond global estimates, the system identifies specific stress points within the distribution network before they reach a breaking point.

- **Ensemble Modeling:** Leverages Gradient Boosting (XGBoost) to capture complex, non-linear consumption patterns and seasonal shifts.
- **Feeder-Level Granularity:** Aggregates individual meter data to generate high-fidelity load curves for specific distribution transformers.
- **Proactive Grid Readiness:** Forecasts short-term surges (hourly and day-ahead), providing operators with a preemptive "Stress Warning" to prevent hardware failure.

Zone-Level Stress Classification & Grid Risk Mapping

Static thresholds are replaced with dynamic risk mapping to visualize the health of the entire grid. By categorizing geographic zones based on real-time and predicted load ratios, the system provides an immediate tactical overview for load-balancing decisions.

- **Multivariate Tiering:** Zones are automatically classified into four operational tiers based on the ratio of predicted demand to historical capacity:
 - **Low (Green):** Stable operations with significant headroom.
 - **Medium (Blue):** Normal peak activity; routine monitoring required.
 - **High (Amber):** Approaching capacity; proactive load-shifting recommended.
 - **Critical (Red):** Threshold exceeded; immediate risk of feeder trip or transformer stress.
- **Geospatial Intelligence:** Translates numerical predictions into a visual heatmap, allowing operators to identify "hotspots" across the distribution network at a glance.
- **Prioritized Alerting:** The system automatically pushes high-risk zone data to the top of the operational queue, ensuring that critical areas receive immediate technical attention.

REVENUE PROTECTION & ANOMALY DETECTION (PART B)

Our detection framework utilizes a multi-model ensemble to identify Non-Technical Losses (NTL) with high precision. By combining traditional statistical outlier detection with neural reconstruction analysis, the system distinguishes between legitimate usage swings and malicious interference.

Advanced Detection Technologies

- **Isolation Forest (Scikit-Learn):** An unsupervised ensemble algorithm that "isolates" anomalies by randomly partitioning data. It is highly efficient at catching global outliers and sudden, drastic consumption shifts.
- **Neural Autoencoders (MLP Architecture):** A Deep Learning approach where a model is trained to reconstruct "normal" usage patterns. A high Reconstruction Error at inference time signals subtle, complex anomalies that standard statistical methods often miss.
- **Ensemble Scoring Logic:** A weighted fusion of both Isolation Forest scores and Autoencoder residuals, providing a unified "Suspicion Index" for every meter.
- **Heuristic Classification Layer:** A rule-based post-processor that categorizes flags into actionable types: Theft (sustained drops), Tamper (erratic spikes), and Peer Deviation (unauthorized consumption vs. neighbors).

THE "GLASS BOX" PRINCIPLE

Auditable Intelligence & The Trust Layer: The GridSense AI Explainability Engine is the critical architectural bridge that ensures AI transparency. It serves as a verification layer that prevents algorithmic bias and provides BESCOM officials with the evidentiary trail required for legal or administrative action following a theft discovery.

Advanced Detection Technologies

- **SHAP Values:** Utilizes SHAP (SHapley Additive exPlanations) to deconstruct model decisions. It identifies exactly which features such as a 45% sudden drop vs. a 5% peer increase triggered the flag.
- **Confidence Scorer:** Converts raw machine learning residuals into a standardized 0–1 Confidence Score, allowing BESCOM to prioritize high-probability theft cases and optimize field-crew dispatch.
- **Noise Reduction:** A dual-stage False Positive Filter uses rule-based guards to distinguish between legitimate variability (e.g., vacations) and suspicious patterns (e.g., partial bypass).
- **Audit Logger:** Every decision, feature weight, and model version is recorded in a persistent JSONL Audit Log, providing a fully reversible and legally defensible record of all system flags.

Operational Efficiency Metrics

Our system translates complex data into high-confidence operational results. By minimizing RMSE in demand forecasting, we provide grid operators with accurate load curves to prevent transformer failures. Simultaneously, we optimize for Recall in loss detection, ensuring that the widest range of theft and tampering signatures are identified for investigation rather than being dismissed as natural variability.

The operational breakthrough lies in our False Positive Mitigation strategy. By cross-referencing flags with peer-group trends and outage logs, we eliminate "noise" and ensure field crews are dispatched only to high-probability sites. This reduces overhead and wasted man-hours, creating a scalable, high-speed architecture ready for city-wide rollout.



SCALABILITY & IMPLEMENTATION ROADMAP

The implementation strategy follows a structured, four-phase rollout designed for low-risk, high-impact scaling. This phased approach ensures AI models are rigorously validated against live operational data before full-scale deployment, maintaining strict adherence to data residency and technical constraints at every milestone.



Phase 1: Controlled Pilot (0–6 Months)

Deployment across a representative sample of 1,000 meters. This phase focuses on calibrating the False Positive Filter and validating the Demand Forecasting RMSE against actual feeder stress events.



Phase 2: Regional Integration (6–18 Months)

Scaling to a full sub-division (10,000+ meters). This involves integrating the pipeline with Kafka-based data streams for real-time processing and deploying the geospatial risk heatmap for regional operators.



Phase 3: BESCOM-Wide Rollout (18–36 Months)

System-wide deployment across millions of meters. During this phase, the Explainability Engine becomes the primary decision-support tool for all field-level inspections and revenue recovery operations.



Phase 4: Grid Edge Evolution (36+ Months)

Transitioning from reactive detection to Proactive Resilience. Integrating external variables—such as EV charging loads and distributed solar—to refine the demand forecasting model for a modernized, green grid.

THANK YOU!
